



Delineation of Groundwater Aquifer and Subsurface Structures of Some Selected Areas in Jema'a Local Government Area of Kaduna State, Nigeria using Electrical Resistivity

 **Jacob J. K.¹**;  **Ruth K.²**;  **Anna A.¹**; &  **Solomon K. D.³**

¹Department of Physics, Kaduna State College of Education Gidan Waya. ²Department of Integrated Science, Kaduna State College of Education Gidan Waya. ³Department of Primary Education Studies, Kaduna State College of Education Gidan Waya

Corresponding Author: megarba7@gmail.com

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Abstract

This study aims to identify shallow groundwater Aquifer and Subsurface Structures of Some Selected Areas in Jema'a Local Government Area of Kaduna State, Nigeria using Electrical Resistivity. Eighteen vertical electrical sounding (VES) surveys were conducted using the Schlumberger array and the apparent resistivity from the obtained VES measurements were interpreted using curve matching inversion to obtain depth, thickness, and resistivity of various subsurface layers across the study area of interest which were eventually used to deduce the lithology and subsurface conditions of the area. The Vertical Electrical Sounding (VES) results revealed variations in the subsurface geo-electric characteristics across the study area, with the interpreted curves showing different geo-electric configurations, including H-, A-, and K-type curves. The H-type curves, characterized by a resistive–conductive–resistive sequence indicate the presence of relatively conductive intermediate layers that may represent weathered and/or water-saturated formations and therefore constitute potential aquifers zones. The A-type curve, characterized by progressively increasing resistivity with depth suggesting a transition from relatively conductive near-surface materials to more resistive weathered or fractured materials and ultimately competent basement, with groundwater occurrence potentially associated with the weathered or fractured zone overlying the basement. The K-type curve indicated a relatively

resistive intermediate layer underlain by a more conductive formation, which may represent a deeper saturated or fractured zone and thus a potential groundwater-bearing horizon. Overall, the VES results suggested that groundwater occurrence within the study area is mainly controlled by the presence, thickness and degree of weathering and fracturing of subsurface. Based on the VES results, it is recommended that groundwater development should focus on locations characterized by relatively thick and moderately conductive weathered and/or fractured formations, which constitute the most promising aquifers zones within the study area. The identified low-resistivity zones should however, be validated through borehole drilling and lithological logging because conductive responses may also be associated with clay-rich materials. Integration of VES with ERT, geological mapping and structural information is recommended to improve the delineation of groundwater-bearing formations.

Keywords: Delineation, Aquifer, groundwater, Resistivity.

Introduction

Jema'a is one of the 23 local government areas of kaduna state. The local government is situated in the southern part of the state, with kafanchan as its headquarters. This local government has some social amenities such as state university and college of education that have over the years, attracted people from other places to settle there either as students, workers or traders. This development has led to the drastic increase of population of the local government in recent years. This increase in population is associated with increase in demands of clean and quality water for sustainable development. The escalating demand for potable and dependable water supplies has underscored the critical importance of groundwater investigations, particularly within rapidly urbanizing environments. Groundwater serves as an indispensable resource for domestic, agricultural, and industrial applications, especially in regions where surface water is either insufficient or compromised by contamination. Accordingly, groundwater exploration has emerged as a strategic intervention to address the intensifying water needs of Nigerian cities and

semi-urban settlements (Beabu , Onengiyeofori and Opiriyabo 2024). This groundwater is the water stored within saturated subsurface voids which is contained in geologic formations known as aquifers. Recharge occurs through rainfall infiltration, with exploration feasible where rocks in the saturation zone exhibit adequate permeability. The increasing demand for potable water reflects population growth across ecosystems. Basher and Main (2014) emphasize groundwater exploration as a vital natural resource for sustaining life, while Utom et al. (2012) highlight the role of water availability in the advancement of civilizations. Inadequate supply of water constrains social welfare and economic development; a situation that could hamper any meaningful development. This is so because water is life and without it, there would be no life on earth. Water, being indispensable to man, embarking on researches to discover clean and reliable water bearing formation known as aquifer becomes necessary in the field of geoscience. This is because the water bearing formations cannot be everywhere that can just be

seen with necked eyes. To uncover it, geological tools and methods are used in order to view beyond the earth's surface. Geological tools have been successfully used in the field of geoscience in solving groundwater, mining, civil engineering, and pollution problems (Sikah et al., 2016). Different geophysical tools such as geo-electrical, electromagnetic, ground penetrating radar (GPR), gravity, magnetic and seismic are routinely used to delineate subsurface structures and stratigraphic sequences, and for groundwater and ore mineral exploration. Researchers such as Saidat et al, (2026) applied vertical electrical sounding (VES) to delineate potential Groundwater formation in Zaria for Water Safety Planning. Their findings revealed that the subsurface was characterized by a crystalline basement complex, in which water retention depends on the thickness of the weathered regolith and fractured bedrock. In a similar research, Eni et al., (2024) applied electrical resistivity survey methods using the Schlumberger configuration to investigate subsurface formations and assess groundwater vulnerability in Nsukka East and West Local Government Area, Enugu State, Nigeria. Their result revealed five geoelectric layers with the fourth layer delineated as the aquifer layers. The values of resistivity and thickness were used in estimating the geohydraulic which help in evaluating the study area.

In this work, the geoelectrical method is used to delineate the geometry of an important groundwater aquifer in Jema'a local government area of Kaduna state. The principal reason for the growing interest in using geophysical methods for hydrogeological studies is that geophysics provides spatially distributed models of physical properties in regions that are difficult to sample using conventional hydrological methods (Butler 2005). The geophysical models often reveal more detail compared with models derived from hydrogeological data, such as pump tests and observations of hydraulic heads. Furthermore, geophysical methods are less invasive and comparatively inexpensive. Therefore, geophysical surveys can improve hydrogeological characterization provided the geophysical and hydrogeological information is properly interpreted. Magnetic rocks contain spatially variable amounts of induced and remnant magnetizations that perturb the Earth's primary field (Reynolds et al. 1990). In this study vertical electrical sounding (VES) investigations were carried out within the area and the research objective is to detect the suitable location for any construction in the study area.

The Geology of the Area

Situated in southern Kaduna State, Jema'a Local Government Area forms part of Nigeria's Precambrian Basement Complex. The geology is dominated by weathered crystalline rocks, older granites, and pegmatite veins. This setting supports notable occurrences of multi-metallic mineralization, a thick lateritic regolith, and discrete basement aquifers. The Local Government Area is one of the 23 administrative divisions of Kaduna State in northwestern Nigeria. The area has an average elevation of approximately 650 meters above sea level and is characterized by low relief, gently undulating terrain, and scattered low-lying outcrops with occasional ridges of weathered granite. Geographically, Jema'a lies between latitudes 9°11' and 9°27' N, and longitudes 8°00' and 8°17' E. It shares boundaries with Kagoro in Kaura Local Government to the east, Ungwan Rimi in Zangon Kataf Local Government to the north, Jaba Local Government to the west, and Nasarawa State to the south (Garba et al., 2022).

Over the past three decades, the population has grown significantly, reaching about 278,202 according to the 2006 Census. The Local Government encompasses several notable settlements, including Kafanchan, Zikpak, Gidan Waya, Godo-godo, Ambe, Kagoma, Ungwan Baki, Ungwan Fari, Takau among others.

The map of Jema'a Local Government Area showing villages, water ways and road network shown in Figure 1.

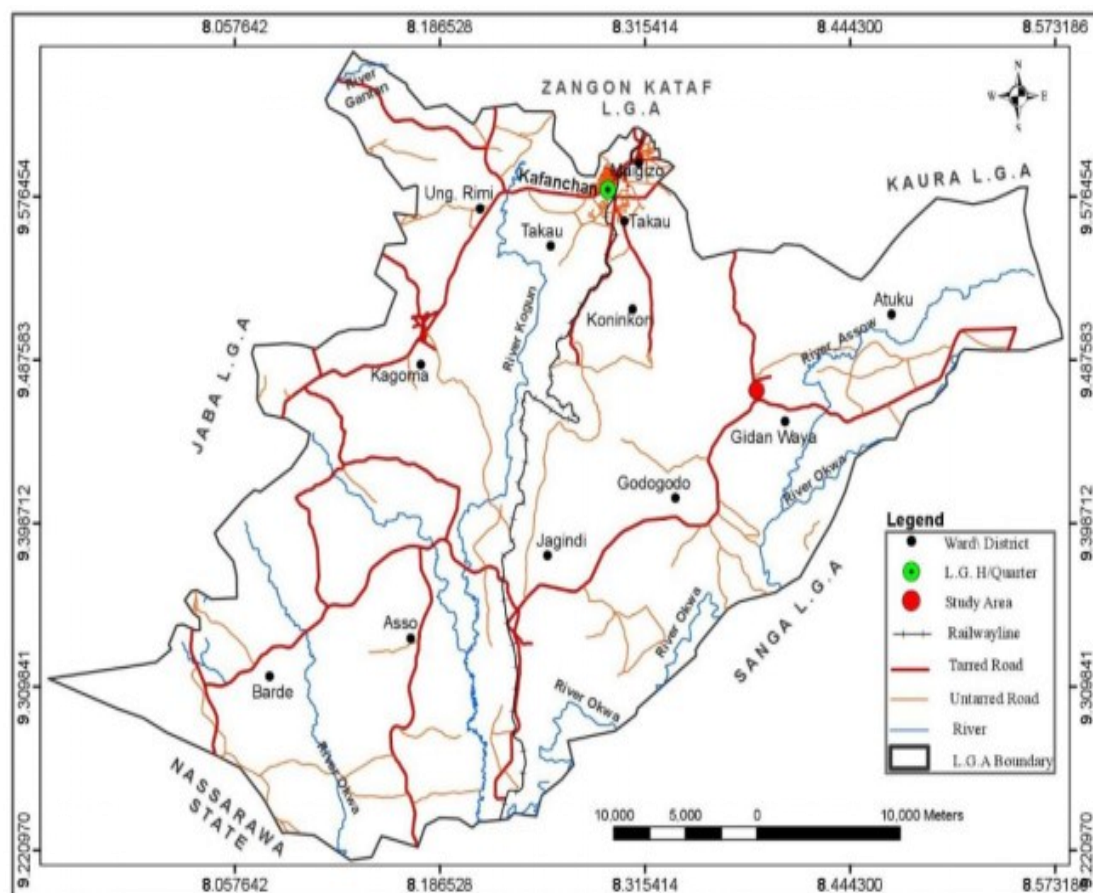


Figure 1: Map of Jema'a Local Government

Materials

The materials for the research survey were Resistivity meter, Electrodes, cables, Power supply, hammer, laptop, GPRS, Basic field tools and processing software.

Method

Water samples were collected from some boreholes of villages across various locations, within the local government area and are listed in Tables I and II. Some boreholes in the study area were functional while others were not. Comparison of data from functional boreholes and dysfunctional ones was made to enable draw some patterns in the geological formations of the study area to conveniently allow delineation of the aquifers zone, since water level and hydraulic conductivity are linked to aquifer and its conductivity. The fact that some were functioning and others were not, signaled a serious failure to reach the water-bearing formation also known as aquiferous zone during the borehole drilling. This experience necessitated some geological survey which led to the choice of electrical resistivity method (vertical electrical soundings), which were carried out at five different villages across selected points within the study area and schlumberger array configuration (figure 2) was employed to collect data.

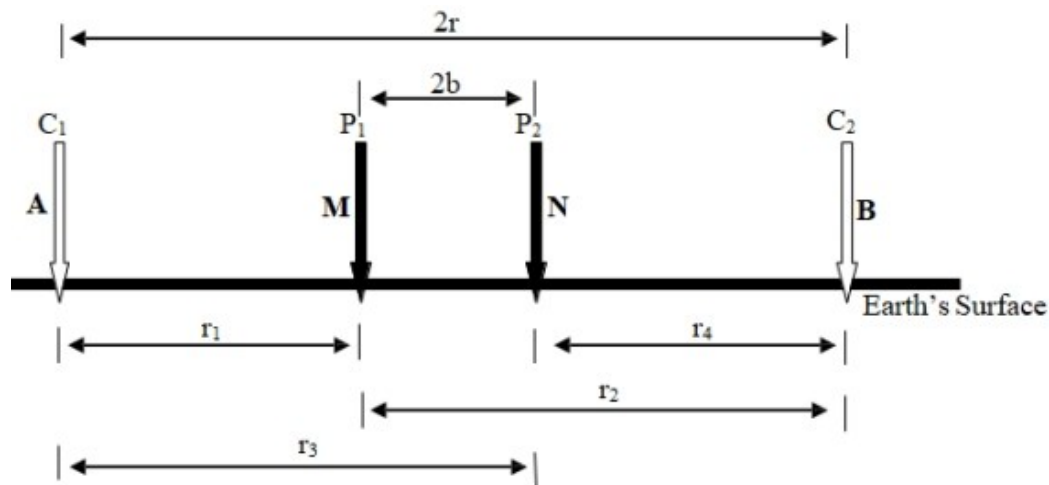


Figure 2: Schlumberger Array

The schlumberger array configuration was used for 2-D imaging since it is moderately sensitive to horizontal and vertical geological structures. This technique was used to aid in understanding the lithology and groundwater exploration of the study area. Seven 2-D stations were carried out with electrodes placed at equidistance of 10m during data collection. To obtain the electrical resistivity values of the subsurface within the study area, an electrical current was injected into the ground through current electrodes and the voltages were measured using the potential electrodes; the geological characteristics of the Earth's subsurface were inferred from these properties. The working principle of this geological process is given in below equations.

Val..... 1

The electrical resistivity method involves the determination of subsurface resistivity distribution by taking ground surface measurements. This requires passing electrical current (I) into the ground by means of two electrodes and the potential difference (ΔV) is measured through another pair of electrodes. The resistivity (ρ) is proportional to the conductor's area and resistance but inversely proportional to its length

$$\rho = \frac{RA}{L} \dots\dots\dots 2$$

This process was employed to avoid the voltage reading from falling below the voltmeter's reading accuracy. The earth is a sphere and the surface area (A) is given

$$A = 2\pi r^2 \dots\dots\dots 3$$

So the change in potential between the points is:

$$\Delta V = \frac{I\rho}{2\pi} \left[\left(\frac{1}{r_1} - \frac{1}{r_2} \right) - \left(\frac{1}{r_3} - \frac{1}{r_4} \right) \right] \dots\dots\dots 3$$

Where r_1, r_2, r_3 and r_4 separations as shown as in figure 4 above

$$\Delta\rho = \frac{2\pi\Delta V}{I} \left[\frac{1}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right) - \left(\frac{1}{r_3} - \frac{1}{r_4}\right)} \right] \dots\dots\dots 4$$

Given that practically the body is inhomogeneous, apparent resistivity (ρ_a) is considered,

$$\rho_a = \frac{2\pi\Delta V}{I} \dots\dots\dots 5$$

Data acquisition for Vertical electrical sounding (VES)

In this survey, a total of twenty four (24) vertical electrical sounding points were obtained with four (4) profiles and six (6) VES points across each profile. The VES spacing was 5m across each profile with the profiles spread (AB) of 200meters. The field procedure involves AB expanding while MN remains fixed and consequently, potential difference decreased across MN, which eventually exceeded the instrument's measuring capability (Arabi, 2009). As a result, a new value of MN is produced which was larger than the previous values, and the survey continued until the last two or three AB values were replicated with the new MN value.

Data acquisition for horizontal profiling

The current electrodes A and B represent the current electrodes through which the current was transferred into the ground through the anode in the schlumberger array that was used. M and N are the potential electrodes across which the current-induced potential difference was measured. All four electrodes A, B, M, and N are positioned on a straight line at equal distance from each other in this setup. A total of four (4) horizontal profiling data were collected along areas containing underground pipes, with an azimuth of NW-SE and a maximum profile length of 150 meters for each profile with a 5meter interval. Equation ($K=2\pi a$) was used to calculate the geometrical factor values. On a log-log graph sheet, the acquired resistivity (ρ) values were plotted against the half current electrode separation (AB/2). On the graph, resistivity (ρ) is on the ordinate, whereas electrode spacing (AB/2) is on the abscissa.

During data acquisition, the researchers took the following into considerations

- a) the electrodes penetrated well into the ground to have good contact by using hammer
- b) connection was carefully made in accordance with schlumberger array.
- c) all the electrodes were connected to the resistivity meter via multicore.

The device was automated such that it selected a pair of electrodes to inject current while measuring the corresponding voltages through another pair of electrodes automatically. It kept repeating the process until all readings for the entire profiles were taken. It was done base on the parameters already programmed by the surveyor. To Process and analyse the resistivity data, the RES2DINV program was used. This program operates effectively because the inversion of quasi-Newton is embedded in the software and makes the analytical calculation of the Jacobian matrix for a homogeneous half-space for the first iteration possible. Also, it helps to interpret the Vertical Electrical Sounding (VES) and to determine the vertical variation in resistivity as a function of depth (Adeka et al., 2025).

Table I. Borehole survey report for the study area

S/N	LON	LAT	ELEVATION (m)	BOREHOLE STATUS	BOREHOLE LOCATION
1	8.057	9.119	628	Non functional	Kafanchan
2	8.257	9.100	619	Functional	Gidan waya
3	8.113	9.128	651	Functional	Godogodo
4	8.202	9.109	646	Non functional	Ambam
5	8.004	9.220	620	Functional	Zikpak
6	8.355	9.019	639	Non functional	Ungwan fari

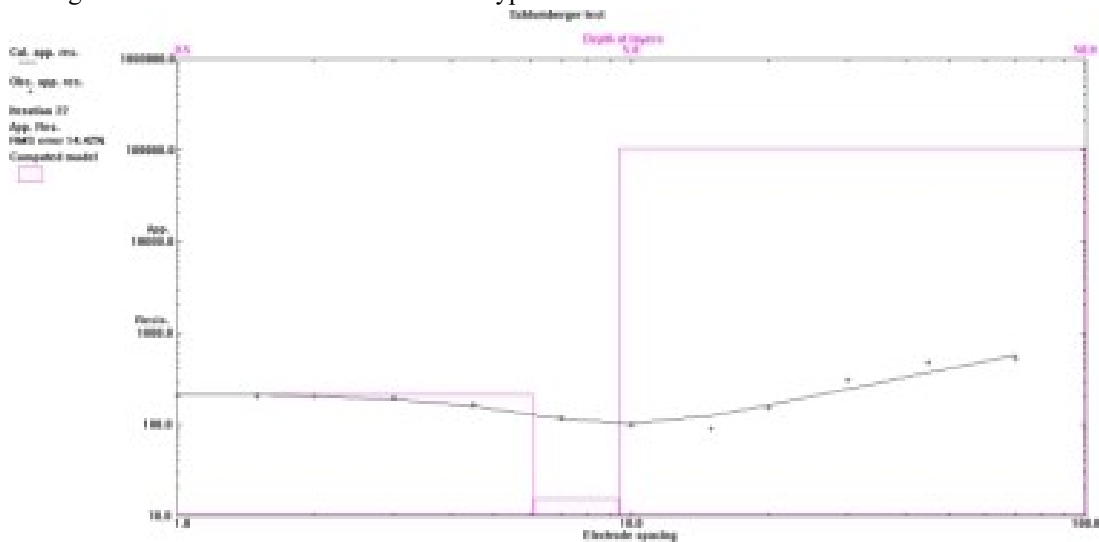
Result and Lithology of Geoelectrical section of profiles

Locations	Layers	Resistivity (Ω m)	Thickness (m)	Depth (m)	Curve Type	Probable Lithology
Kafanchan	1	215.387	2.844	2.844	H-type	Partially weathered soil
	2	34.463	4.480	7.32		Potentially weathered formation; possible aquifer
	3	3403.481				More resistive consolidated formation
Gidan waya	1	204.081	0.96	0.96	H-type	Partially weathered soil
	2	86.319	11.8	12.8		Potential aquiferous zone
	3	2652.3				Fresh basement
Godogodo	1	291.73	0.7	0.7	H-type	Partially weathered zone
	2	61.220	12.7	14.4		Potentially saturated weathered
	3	1445.55				Fresh basement
Ambam	1	66.061	1.357	1.36	H-type	Partially weathered zone
	2	24.269	5.094	6.50		Potential aquiferous zone
	3	2325.011				Fresh basement
Zikpak	1	229.499	0.543	0.540	K-type	Partially weathered zone
	2	333.177	65.359	65.900		Fresh basement
	3	232.063				Potential aquiferous zone
U/Fari	1	260.842	2.455	2.455	A-type	Weathered zone
	2	268.864	17.341	18.80		More fractured zone
	3	3448.468				Fresh basement rock

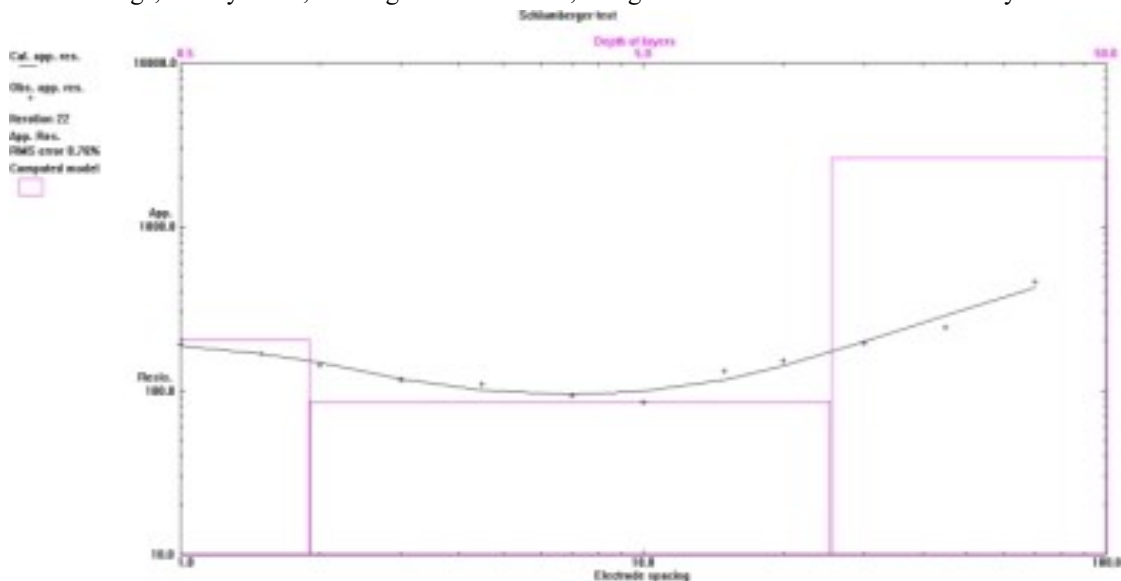
RESULTS AND DISCUSSION

Layer	Resistivity	Thickness	Depth
1	215.387	2.844	2.844
2	34.463	4.480	7.32
3	3403.481		

The results gotten from VES (using schlumberger array) of the six (6) stations were presented in Tables I and II. From this, the resistivity values, thickness and their corresponding depths across the study area were given and written in terms of the curve types.

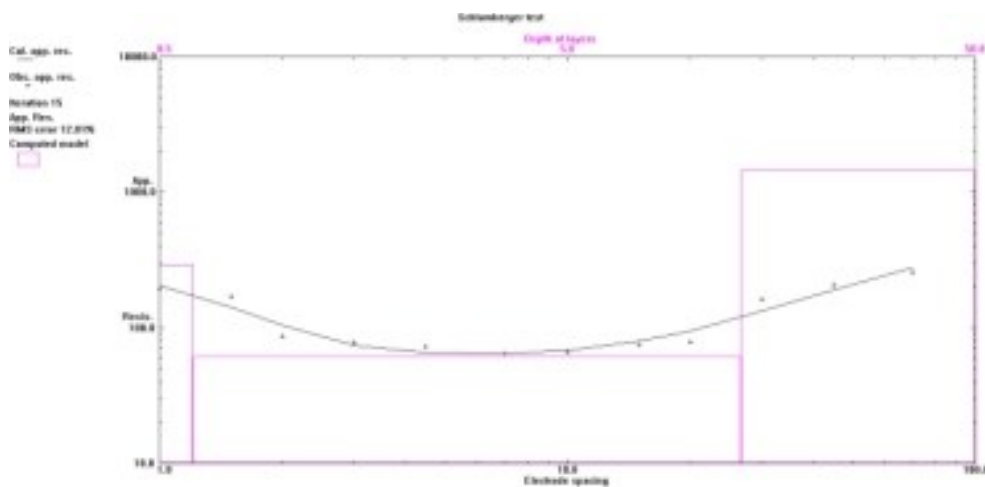


From the results shown above, the second geoelectric layer (approximately 7.3m) is the most significant zone for groundwater investigation. Its relatively lower resistivity compared with the overlying and underlying layers produces the characteristic H-type response ($\rho_1 > \rho_2 < \rho_3$). This lower-resistivity interval may represent a water-bearing formation, particularly if the local geology consists of permeable sand, sandstone, weathered material, or fractured rock. Groundwater increases the electrical conductivity of a formation and commonly results in lower resistivity than the corresponding dry formation. However, an important qualification is necessary: low resistivity alone does not prove the presence of groundwater. Clay-rich formations can also have very low resistivity. Therefore, the second layer should be regarded as a probable aquifer zone, and its interpretation should be correlated with borehole logs, nearby wells, lithological information, and groundwater occurrence in the study area.

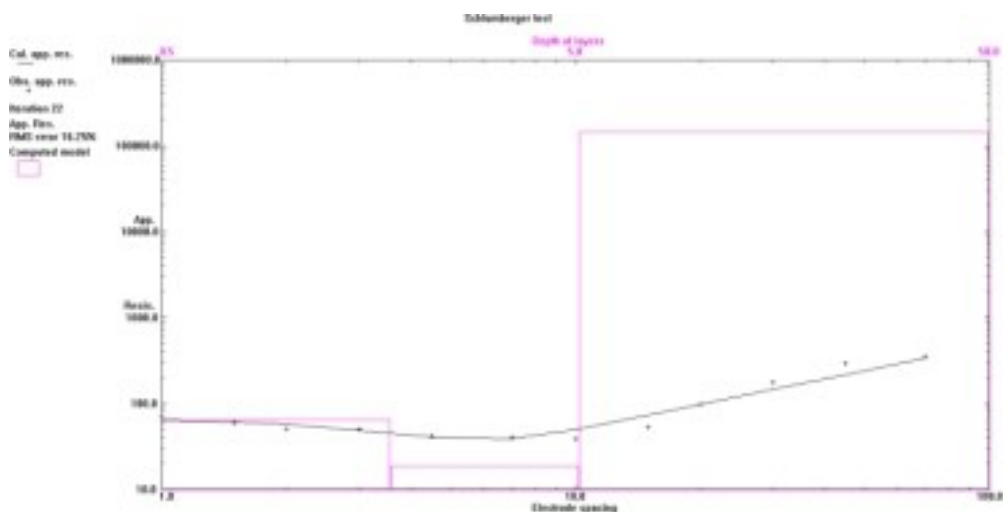


Layer	Resistivity	Thickness	Depth
1	204.081	0.96	0.96
2	86.319	11.8	12.8
3	2652.3		

The H-type VES curve ($\rho_1 > \rho_2 < \rho_3$) indicates a three-layer subsurface sequence. The intermediate low-resistivity layer at a depth of about 12.8m is interpreted as a potential aquifers zone, possibly corresponding to a saturated weathered/fractured formation. The relatively high-resistivity underlying layer may represent fresh basement, although fractured portions of the basement may also constitute groundwater-bearing zones. The interpretation should be constrained with geological and borehole information because low resistivity may also result from clay-rich materials.



Layer	Resistivity	Thickness	Depth
1	291.733	0.7	0.7
2	61.220	12.7	14.4
3	1445.5		

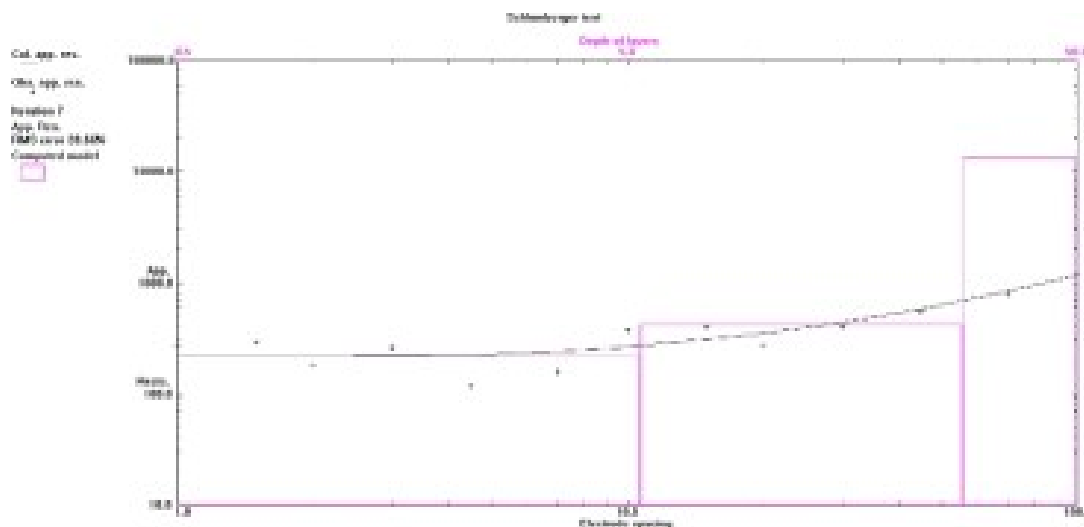


The VES curve exhibits an H-type geoelectric configuration ($\rho_1 > \rho_2 < \rho_3$), indicating a three-layer subsurface sequence. The first layer is characterized by relatively high resistivity and is interpreted as topsoil/lateritic material. The second layer has relatively low resistivity and is interpreted as a potentially saturated weathered or fractured zone. This layer at a depth of 14.4m constitutes the main target for groundwater exploration and may represent aquifers horizon. The third, relatively high-resistivity layer is interpreted as fresh or competent basement. However, the possibility of clay-rich material producing low resistivity cannot be excluded; hence, geological and/or borehole information is required to confirm the aquifer interpretation.

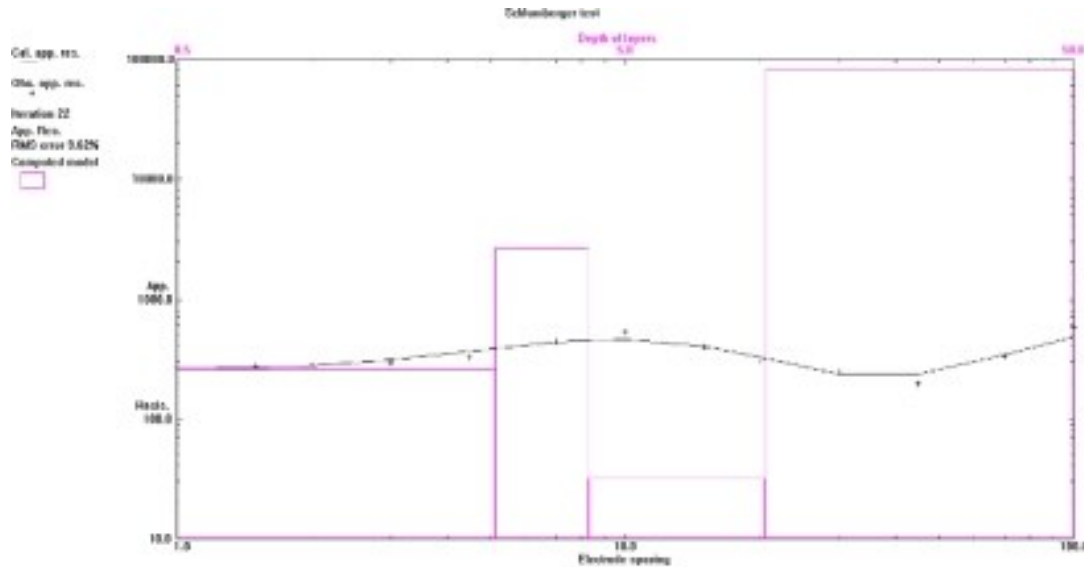
Layer	Resistivity	Thickness	Depth
1	66.061	1.357	1.36
2	24.269	5.094	6.5
3	2325.011		

Layer	Resistivity	Thickness	Depth
1	260.842	2.455	2.455
2	268.864	17.341	19.8
3	3448.462		

The VES curve displays an H-type geo-electric configuration ($\rho_1 > \rho_2 < \rho_3$), suggesting a three-layer subsurface sequence. The upper layer is characterized by relatively high resistivity and is interpreted as topsoil/lateritic material. The second layer exhibits relatively low resistivity at a depth of about 6.5m and is interpreted as a weathered and potentially water-saturated zone. This layer constitutes the principal aquifer target at the VES station. The underlying high-resistivity layer is interpreted as relatively fresh basement rock. Nevertheless, the low resistivity of the second layer may also be associated with clay-rich materials; therefore, confirmation using borehole lithology, nearby well information, or other geophysical evidence is recommended.”



The VES curve exhibits an approximately A-type geoelectric configuration ($\rho_1 < \rho_2 < \rho_3$), characterized by a progressive increase in resistivity with electrode spacing. The first layer is interpreted as the near-surface weathered/topsoil material, while the second layer represents a relatively more resistive weathered or fractured formation that constitutes a potential aquiferous horizon. The third layer is characterized by high resistivity and is interpreted as relatively fresh basement rock. Groundwater potential at this location is therefore associated primarily with the weathered/fractured zone above the fresh basement. Confirmation of the aquifers nature of the intermediate layer requires correlation with borehole, well, or geological information.



Layer	Resistivity	Thickness	Depth
1	229.499	0.543	0.54
2	333.177	65.359	65.9
3	232.063		

The VES curve exhibits a K-type geoelectric configuration ($\rho_1 < \rho_2 > \rho_3$), indicating a three-layer subsurface sequence in which the intermediate layer has a higher resistivity than the overlying and underlying layers. The first layer is interpreted as topsoil or near-surface weathered material. The relatively resistive second layer may represent a less-weathered or more competent formation, while the underlying low-resistivity third layer may indicate a saturated fractured/weathered zone and therefore constitutes a potential aquiferous horizon. However, the low resistivity may also be associated with clay-rich materials; hence, correlation with borehole and geological information is necessary to confirm its groundwater potential.

DISCUSSION OF RESULTS

The Vertical Electrical Sounding (VES) results revealed variations in the subsurface geo-electric characteristics across the study area, with the interpreted curves showing different geo-electric

configurations, including H-, A-, and K-type curves. The H-type curves, characterized by a resistive–conductive–resistive sequence ($\rho_1 > \rho_2 < \rho_3$), indicate the presence of relatively conductive intermediate layers that may represent weathered and/or water-saturated formations and therefore constitute potential aquifers zones. The A-type curve, characterized by progressively increasing resistivity with depth ($\rho_1 < \rho_2 < \rho_3$), suggesting a transition from relatively conductive near-surface materials to more resistive weathered or fractured materials and ultimately competent basement, with groundwater occurrence potentially associated with the weathered or fractured zone overlying the basement. The K-type curve ($\rho_1 < \rho_2 > \rho_3$) indicated a relatively resistive intermediate layer underlain by a more conductive formation, which may represent a deeper saturated or fractured zone and thus a potential groundwater-bearing horizon. Overall, the VES results suggested that groundwater occurrence within the study area is mainly controlled by the presence, thickness and degree of weathering and fracturing of subsurface.

Recommendations

Based on the VES results, it is recommended that groundwater development should focus on locations characterized by relatively thick and moderately conductive weathered and/or fractured formations, which constitute the most promising aquifers zones within the study area. The identified low-resistivity zones should however, be validated through borehole drilling and lithological logging because conductive responses may also be associated with clay-rich materials. Integration of VES with ERT, geological mapping and structural information is recommended to improve the delineation of groundwater-bearing formations. Prior to large-scale groundwater development, pumping tests and comprehensive water-quality analyses should be undertaken to determine aquifer productivity and groundwater suitability.

Acknowledgement

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